ChatGPT

Study/Paper Forensics Expansion Pack v0.2

Executive summary

This v0.2 expansion pack extends the initial six-case pilot with eight carefully curated cases that probe additional epistemic failure modes and broaden the domain coverage beyond health and supplements. The new cases include solid real trials, overclaimed AI and environmental reports, a mechanism-heavy gene therapy summary, correlation–causation traps, and synthetic contrived examples. Two cases are labelled **borderline**, acknowledging ambiguity where strong analysts might disagree. Together, these cases strengthen the dataset's ability to test Agent Arena's handling of educational, workplace, environmental, longevity and technology claims while maintaining high signal and manageable length. Each case is short, paraphrased to avoid copyright issues, and stripped of citation markers. They are designed to expose overclaim, mechanism–evidence tension, unsupported generalisation and correlation masquerading as causation, while also including clear solid examples to guard against overflagging.

Dataset table

case_id	title	bucket	domain	source_type	gold_case_type	expected_
Case021	Virtual tutoring boosts reading skills	solid	education	real_summary	clear	none_or_r
Case022	Four-day workweek trial improves well-being	solid	workplace/ economics	real_summary	borderline	none_or_r
Case023	Urban green space and mental health	correlation_vs_causation	environment/ public health	real_summary	clear	correlatio
Case024	Gene therapy elevating Klotho protein extends lifespan in mice	mechanism_vs_evidence	longevity/ gene therapy	real_summary	clear	mechanis

case_id	title	bucket	domain	source_type	gold_case_type	expected_
Case025	AI instruction outperforms human teachers	overclaim	education/AI	real_summary	clear	unsuppor
Case026	Gas stoves and childhood asthma risk	correlation_vs_causation	environment/ public health	real_summary	borderline	correlatio
Case027	Smartwatch app claims to raise IQ	overclaim	technology/ cognitive enhancement	synthetic_case	clear	unsuppor
Case028	Office plants and employee productivity	correlation_vs_causation	workplace/ productivity	synthetic_case	clear	correlatio

Full case definitions

Case021 – Virtual tutoring boosts reading skills

```
case_id: Case021
title: Virtual tutoring boosts reading skills
domain: education
category_bucket: solid
difficulty: medium
source_type: real_summary
expected_strength: strong
expected_main_failure: none_or_minor

# Input
input_text: |
  A randomized controlled trial tested a virtual reading tutoring program in
a rural school district.  The program paired live online tutors with small
groups of struggling elementary students for 10-week sessions.  Students who
completed around forty tutoring sessions gained roughly two additional months
of reading progress compared with peers in the control condition.  Even those
attending fewer sessions showed modest gains.  Researchers emphasised that
consistent participation was critical and highlighted that the program
delivered results at a lower cost than typical high-dose tutoring.

# Gold layer
gold_label: solid
gold_confidence: high
gold_case_type: clear
gold_summary: The summary describes a rigorously designed trial of a virtual
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tutoring program.  It reports improved reading outcomes and notes cost
effectiveness and the importance of consistent participation without
overstating broader impacts.
gold_justification:
A randomized controlled trial with clear outcome measures supports the
claims.  The summary uses cautious language (“gained roughly two additional
months”, “consistency is critical”) and does not generalise beyond the study
population.  There is no obvious mechanism–evidence mismatch or causal
overreach.
gold_evidence_span: Students who completed around forty tutoring sessions
gained roughly two additional months of reading progress compared with peers
in the control condition.
gold_alternative_reading: null

# Expected findings
expected_findings:
  - finding: evidence_matches_claim
    severity: minor
    confidence: high
  - finding: cautious_scope_control
    severity: minor
    confidence: high

# Evidence spans
evidence_spans:
  - span: students who completed around forty tutoring sessions gained
roughly two additional months of reading progress
    why_it_matters: Shows that the primary claim (extra reading progress) is
directly supported by measured outcomes in the trial.
    supports_finding: evidence_matches_claim
  - span: researchers emphasised that consistent participation was critical
and highlighted that the program delivered results at a lower cost than
typical high-dose tutoring
    why_it_matters: Demonstrates cautious language and avoids
overgeneralization by noting conditions for success and cost context.
    supports_finding: cautious_scope_control

# Diagnostic value
why_this_case_is_useful: This case provides a clean, positive example of a
well-designed educational RCT.  It helps calibrate Agent Arena to avoid
flagging non-problems and distinguishes genuine evidence-bound claims from
hype.  The short summary still mentions limitations and avoids causal
overreach, serving as a control against overflagging.

# Optional variant
variant_type: overclaim_summary
variant_text: |
  A press article trumpeted that a new virtual tutoring program
“revolutionises education” after a trial in a rural district.  It claimed
that students using the program jumped ahead by two grade levels in reading
and that the technology would “close the achievement gap nationwide” while
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saving schools thousands of dollars.  It described the program as a silver
bullet for literacy without discussing dosage, study population or the need
for teacher involvement.
```

Case022 – Four-day workweek trial improves well-being

```
case_id: Case022
title: Four-day workweek trial improves well-being
domain: workplace/economics
category_bucket: solid
difficulty: medium
source_type: real_summary
expected_strength: medium
expected_main_failure: none_or_minor

# Input
input_text: |

A six-month experiment across 141 companies in six countries tested whether
cutting the work week from five days to four while maintaining pay could
improve employee well-being.  Companies volunteered to restructure their
workflows so that productivity would not decline with fewer hours.  Nearly
2,900 participants completed surveys before and after the trial.  After six
months, workers reported lower stress and burnout, higher job satisfaction,
and no perceived drop in performance.  More than 90 percent of participating
companies chose to continue the four-day schedule.  The researchers cautioned
that because companies self-selected into the trial and outcomes were
self-reported, further randomised evaluations are needed.

# Gold layer
gold_label: solid
gold_confidence: medium
gold_case_type: borderline
gold_summary:
The summary reports results from a large multi-company trial of a four-day
workweek showing improvements in employee well-being without apparent
productivity loss.  It notes limitations such as voluntary participation and
self-reported outcomes.
gold_justification: Although the study was not randomised and relies on
self-report, the summary accurately conveys the evidence and includes
caveats.  The claims are restrained to the measured outcomes, and the need
for further research is acknowledged, so it does not constitute overclaim.
gold_evidence_span: After six months, workers reported lower stress and
burnout, higher job satisfaction, and no perceived drop in performance.
gold_alternative_reading: Some may argue that self-selection and
self-reported measures limit the evidence quality, making the case less
solid.

# Expected findings
expected_findings:
```

```
  - finding: evidence_matches_claim
    severity: minor
    confidence: medium
  - finding: honest_uncertainty
    severity: minor
    confidence: medium

# Evidence spans
evidence_spans:
  - span:
workers reported lower stress and burnout, higher job satisfaction, and no
perceived drop in performance
    why_it_matters: Directly supports the claim that the four-day schedule
improved well-being without harming productivity.
    supports_finding: evidence_matches_claim
  - span:
researchers cautioned that because companies self-selected into the trial and
outcomes were self-reported, further randomised evaluations are needed
    why_it_matters: Provides explicit acknowledgement of methodological
limitations, representing honest uncertainty.
    supports_finding: honest_uncertainty

# Diagnostic value
why_this_case_is_useful: This borderline case evaluates whether Agent Arena
can recognise cautious yet positive findings without overflagging.  It
includes explicit limitations (self-selection, self-report) and invites
debate on strength of evidence.  It broadens the dataset beyond health into
workplace/economic policy.

# Optional variant
variant_type: press_style_summary
variant_text: |
  Headlines proclaimed that switching to a four-day workweek “made workers
happier and more productive” after a six-month trial.  Reports emphasised
that 90 percent of firms kept the new schedule and suggested that the policy
could solve burnout globally.  Many articles did not mention that companies
volunteered for the trial or that outcomes were self-reported, implying a
causal link between reduced hours and productivity for all workplaces.
```

Case023 – Urban green space and mental health

```
case_id: Case023
title: Urban green space and mental health
domain: environment/public health
category_bucket: correlation_vs_causation
difficulty: medium
source_type: real_summary
expected_strength: weak
expected_main_failure: correlation_as_causation
```

```yaml
# Input
input_text: |
  Researchers used a neighbourhood NatureScore to measure the amount and
quality of green space across over a thousand ZIP codes in urban Texas.  They
linked these scores to aggregated outpatient records for depression, bipolar
disorder, stress and anxiety among more than 61 million encounters.
Neighbourhoods with higher NatureScores had about 50 percent fewer mental
health visits, and residents living in areas scoring above 40 points were
reported to have a 51 percent lower likelihood of depression and a 63 percent
lower likelihood of bipolar disorder.  The authors suggested that increasing
green space in cities could promote well-being and improve mental health
outcomes.

# Gold layer
gold_label: correlation_vs_causation
gold_confidence: high
gold_case_type: clear
gold_summary: The summary describes a cross-sectional analysis linking
green-space scores to mental health service utilisation.  It reports strong
associations and speculates about benefits of increasing urban greenness but
does not establish causation.
gold_justification: The study measured correlations between green space and
mental health at the ZIP-code level.  Without experimental or longitudinal
design, one cannot infer that green space causes better mental health.
Socioeconomic differences could confound the observed associations.
gold_evidence_span: Neighbourhoods with higher NatureScores had about 50
percent fewer mental health visits, and residents living in areas scoring
above 40 points were reported to have a 51 percent lower likelihood of
depression and a 63 percent lower likelihood of bipolar disorder.
gold_alternative_reading: null

# Expected findings
expected_findings:
  - finding: correlation_as_causation
    severity: major
    confidence: high
  - finding: scope_overreach
    severity: minor
    confidence: medium

# Evidence spans
evidence_spans:
  - span: Neighbourhoods with higher NatureScores had about 50 percent fewer
mental health visits
    why_it_matters:
Shows the strength of the reported association, which may be mistaken for
causation.
    supports_finding: correlation_as_causation
  - span: residents living in areas scoring above 40 points were reported to
have a 51 percent lower likelihood of depression and a 63 percent lower
likelihood of bipolar disorder
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    why_it_matters:
Presents impressive percentage reductions that could lead to
overinterpretation of causality.
    supports_finding: correlation_as_causation
  - span: The authors suggested that increasing green space in cities could
promote well-being and improve mental health outcomes
    why_it_matters: Indicates scope overreach—recommendations for policy
changes based on correlational data.
    supports_finding: scope_overreach

# Diagnostic value
why_this_case_is_useful: This case tests whether Agent Arena recognises
correlation-causation traps.  The summary emphasises striking associations
and policy suggestions without establishing a causal mechanism.  It helps
evaluate the system's ability to flag causal overreach while considering that
the researchers used large data sets and credible methods.
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Case024 – Gene therapy elevating Klotho protein extends lifespan in mice

```
case_id: Case024
title: Gene therapy elevating Klotho protein extends lifespan in mice
domain: longevity/gene therapy
category_bucket: mechanism_vs_evidence
difficulty: hard
source_type: real_summary
expected_strength: medium
expected_main_failure: mechanism_not_demonstrated

# Input
input_text: |
  An international team used viral vectors to boost production of the
secreted form of the Klotho protein in middle-aged mice.  The treated mice
lived 15 to 20 percent longer and showed improvements in muscle strength,
bone density and cognitive performance.  Researchers observed increased
formation of neurons and enhanced immune activity in the hippocampus.  They
filed patents for using Klotho to treat bone, muscle and aging-related
deficits and suggested that with a suitable delivery method, this gene
therapy could help build a healthier society by combating age-related
decline.

# Gold layer
gold_label: mechanism_vs_evidence
gold_confidence: high
gold_case_type: clear
gold_summary: The summary describes a gene-therapy intervention that
increases Klotho protein levels in mice and reports extended lifespan and
improved physiological and cognitive outcomes.  It suggests potential human
applications without presenting human evidence.
gold_justification: While the study demonstrates a mechanism (increased
Klotho) and measurable effects in mice, the article extends the implications
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to human aging.  The mechanism is described but not demonstrated in humans;
translation to people remains speculative.
gold_evidence_span:
The treated mice lived 15 to 20 percent longer and showed improvements in
muscle strength, bone density and cognitive performance.
gold_alternative_reading: null

# Expected findings
expected_findings:
  - finding: mechanism_not_demonstrated
    severity: major
    confidence: high
  - finding: conclusion_exceeds_results
    severity: medium
    confidence: high

# Evidence spans
evidence_spans:
  - span: The treated mice lived 15 to 20 percent longer and showed
improvements in muscle strength, bone density and cognitive performance
    why_it_matters: Demonstrates the robust effects observed in animals but
not yet in humans.
    supports_finding: mechanism_not_demonstrated
  - span: they filed patents for using Klotho to treat bone, muscle and
aging-related deficits and suggested that with a suitable delivery method,
this gene therapy could help build a healthier society by combating
age-related decline
    why_it_matters: Illustrates overextension of animal findings to broad
societal benefits without human evidence.
    supports_finding: conclusion_exceeds_results

# Diagnostic value
why_this_case_is_useful: This case probes the model's ability to detect
mechanism-versus-evidence tension.  It includes detailed description of a
biological mechanism and robust animal data but leaps toward human
applications.  Agent Arena should flag that the mechanism is named and
partially demonstrated but not proven in humans, and caution against
overstated societal implications.
```

Case025 – AI instruction outperforms human teachers

```
case_id: Case025
title: AI instruction outperforms human teachers
domain: education/AI
category_bucket: overclaim
difficulty: medium
source_type: real_summary
expected_strength: weak
expected_main_failure: unsupported_generalization
```

```
# Input
input_text: |
  An opinionated business article argued that specialized AI systems are
rapidly making human teachers obsolete.  Citing pilot programs in the United
States and United Kingdom, it claimed that students taught by AI outperform
peers in physics, mathematics and literacy, with some reports suggesting a
65 percent rise in test scores.  The article described a “2× learning
effect,” where an AI tutor purportedly taught material twice as effectively
and in less time than active-learning classrooms.  It concluded that the role
of educators as the primary source of knowledge is being dismantled and that
AI instruction represents a data-backed reality for all subjects.

# Gold layer
gold_label: overclaim
gold_confidence: high
gold_case_type: clear
gold_summary: The article presents sweeping claims that AI instruction makes
human teachers obsolete and delivers massive learning gains across subjects.
It generalises from limited pilot programs and does not describe study
design, sample size or context.
gold_justification: The narrative extrapolates from isolated pilots to a
universal claim that AI instruction is superior in all core subjects and
implies the end of human teaching.  No evidence is provided to support the
65 percent test-score increase or to show that AI can replace educators; thus
the conclusion exceeds the evidence and constitutes unsupported
generalisation.
gold_evidence_span: Citing pilot programs in the United States and United
Kingdom, it claimed that students taught by AI outperform peers in physics,
mathematics and literacy, with some reports suggesting a 65 percent rise in
test scores.
gold_alternative_reading: null

# Expected findings
expected_findings:
  - finding: conclusion_exceeds_results
    severity: major
    confidence: high
  - finding: unsupported_generalization
    severity: major
    confidence: high

# Evidence spans
evidence_spans:
  - span: students taught by AI outperform peers in physics, mathematics and
literacy, with some reports suggesting a 65 percent rise in test scores
    why_it_matters:
Key overclaim linking AI instruction to large performance gains across
subjects without context.
    supports_finding: unsupported_generalization
  - span: it concluded that the role of educators as the primary source of
knowledge is being dismantled and that AI instruction represents a
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data-backed reality for all subjects
    why_it_matters:
Shows the article's sweeping conclusion that extrapolates from limited
evidence to claim human obsolescence.
    supports_finding: conclusion_exceeds_results

# Diagnostic value
why_this_case_is_useful: This case serves as a strong overclaim example in
the education domain.  It tests whether
Agent Arena can identify when a media narrative inflates preliminary findings
into a broad claim about the future of teaching.  The clear overstatement of
AI's capabilities and lack of methodological detail provide a sharp contrast
to the measured language of solid cases.
```

Case026 – Gas stoves and childhood asthma risk

```
case_id: Case026
title: Gas stoves and childhood asthma risk
domain: environment/public health
category_bucket: correlation_vs_causation
difficulty: medium
source_type: real_summary
expected_strength: weak
expected_main_failure: correlation_as_causation

# Input
input_text: |
  A news story reported that new research links gas stove emissions to
childhood asthma and has prompted advocates to demand the removal of gas
stoves from public housing.  The article stated that nearly 13 percent of
childhood asthma cases are now "attributed" to exposure to gas stovetops and
argued that switching to electric induction stoves could prevent up to 12.7
percent of cases.  It described gas stoves emitting pollutants such as carbon
monoxide, benzene and nitrogen dioxide and framed the issue as an urgent
public health hazard affecting mostly woman-led households in public
housing.  Advocacy groups called for federal agencies to regulate or ban gas
stoves.

# Gold layer
gold_label: correlation_vs_causation
gold_confidence: medium
gold_case_type: borderline
gold_summary: The article summarises a study associating gas-stove exposure
with childhood asthma and presents activist responses.  It uses language like
"attributed" and "prevent" that implies causality, though the underlying
study is observational and may be confounded.
gold_justification: While gas stoves emit pollutants and associations with
asthma exist, attributing a specific percentage of asthma cases to gas stoves
overstates the causal inference from observational data.  The article blends
advocacy and science, making causal claims without controlled experiments.
```

```
gold_evidence_span: The article stated that nearly 13 percent of childhood
asthma cases are now “attributed” to exposure to gas stovetops and argued
that switching to electric induction stoves could prevent up to 12.7 percent
of cases.
gold_alternative_reading: Some may contend that the article is advocating
precaution based on a growing body of evidence, not asserting definitive
causality.

# Expected findings
expected_findings:
  - finding: correlation_as_causation
    severity: major
    confidence: medium
  - finding: conclusion_exceeds_results
    severity: medium
    confidence: medium

# Evidence spans
evidence_spans:
  - span:
nearly 13 percent of childhood asthma cases are now “attributed” to exposure
to gas stovetops
    why_it_matters: Uses causal language (“attributed”) for an observational
association.
    supports_finding: correlation_as_causation
  - span: switching to electric induction stoves could prevent up to 12.7
percent of cases
    why_it_matters: Presents an intervention effect without experimental
evidence, suggesting causality.
    supports_finding: conclusion_exceeds_results

# Diagnostic value
why_this_case_is_useful: This borderline case examines how Agent Arena
handles policy-driven narratives that convert correlational research into
causal policy recommendations.  It challenges the model to distinguish
between advocating precaution and overstating causal certainty.
```

Case027 – Smartwatch app claims to raise IQ (synthetic)

```
case_id: Case027
title: Smartwatch app claims to raise IQ
domain: technology/cognitive enhancement
category_bucket: overclaim
difficulty: easy
source_type: synthetic_case
expected_strength: overclaimed
expected_main_failure: unsupported_generalization

# Input
input_text: |
```

```
  A tech startup announced that its new smartwatch app “NeuroBoost” could
raise users’ IQ by 15 points in just two weeks.  The company said the app
delivers personalised vibrations and colour patterns that “rewire neural
pathways,” leading to substantial gains in working memory and
problem-solving.  According to the founder, a small pilot study of 10
employees showed an average IQ increase from 105 to 120 after using the app
daily for 10 minutes.  The company plans to roll out the app globally and
claims it will revolutionise human intelligence.

# Gold layer
gold_label: overclaim
gold_confidence: high
gold_case_type: clear
gold_summary: This synthetic press-style summary asserts that a smartwatch
app dramatically increases intelligence based on an uncontrolled pilot of 10
employees.  It extrapolates the findings to the general population and uses
pseudo-scientific language about “rewiring neural pathways.”
gold_justification: The claim of raising IQ by 15 points lacks credible
evidence and generalises from a tiny, uncontrolled sample.  The mechanistic
language is not supported by demonstrated neuroscience, and the conclusion
that the app will “revolutionise human intelligence” is an unsupported
generalisation.
gold_evidence_span:
The company said the app delivers personalised vibrations and colour patterns
that “rewire neural pathways,” leading to substantial gains in working memory
and problem-solving.
gold_alternative_reading: null

# Expected findings
expected_findings:
  - finding: unsupported_generalization
    severity: major
    confidence: high
  - finding: mechanism_not_demonstrated
    severity: medium
    confidence: high

# Evidence spans
evidence_spans:
  - span: its new smartwatch app “NeuroBoost” could raise users’ IQ by 15
points in just two weeks
    why_it_matters: Presents the core overclaim of dramatic IQ enhancement.
    supports_finding: unsupported_generalization
  - span: delivers personalised vibrations and colour patterns that “rewire
neural pathways”
    why_it_matters: Introduces a mechanistic claim without evidence.
    supports_finding: mechanism_not_demonstrated
  - span: a small pilot study of 10 employees showed an average IQ increase
from 105 to 120
    why_it_matters: Indicates inadequate evidence and sample size for
sweeping conclusions.
```

```
    supports_finding: unsupported_generalization

# Diagnostic value
why_this_case_is_useful: This synthetic case isolates a clear overclaim with
pseudo-scientific mechanism language.  It challenges Agent Arena to flag
unsupported generalisations and mechanism-not-demonstrated without the
complexity of real-world nuance.  It also introduces a technology/
productivity domain.
```

Case028 – Office plants and employee productivity (synthetic)

```
case_id: Case028
title: Office plants and employee productivity
domain: workplace/productivity
category_bucket: correlation_vs_causation
difficulty: easy
source_type: synthetic_case
expected_strength: weak
expected_main_failure: correlation_as_causation

# Input
input_text: |
  A company's HR newsletter reported that employees who placed two or more
potted plants on their desks had 30 percent higher annual performance ratings
than those without plants.  The newsletter explained that plants create
"positive energy" and concluded that buying desk plants causes employees to
become more productive and innovative.  It encouraged managers to fund indoor
greenery initiatives company-wide, promising substantial returns on
investment.

# Gold layer
gold_label: correlation_vs_causation
gold_confidence: high
gold_case_type: clear
gold_summary: This synthetic case describes an observational association
between desk plants and higher performance evaluations but treats it as
causal, claiming that plants make workers more productive.
gold_justification:
The newsletter infers a causal relationship from a simple correlation without
controlling for confounders (e.g., more conscientious workers might buy
plants).  The claim that plants "cause" higher productivity is unwarranted.
gold_evidence_span: The newsletter explained that plants create "positive
energy" and concluded that buying desk plants causes employees to become more
productive and innovative.
gold_alternative_reading: null

# Expected findings
expected_findings:
  - finding: correlation_as_causation
    severity: major
```

```
    confidence: high
  - finding: unsupported_generalization
    severity: medium
    confidence: medium

# Evidence spans
evidence_spans:
  - span: employees who placed two or more potted plants on their desks had
30 percent higher annual performance ratings than those without plants
    why_it_matters: Shows the correlational evidence used to support the
causal claim.
    supports_finding: correlation_as_causation
  - span: plants create “positive energy” and concluded that buying desk
plants causes employees to become more productive and innovative
    why_it_matters: Explicitly states a causal conclusion from a correlation
and introduces unsupported new-age language.
    supports_finding: unsupported_generalization

# Diagnostic value
why_this_case_is_useful: This synthetic correlation-causation trap tests
whether Agent Arena can spot causal claims based on workplace correlations
and differentiate them from legitimate causal evidence.  It broadens the
dataset into workplace productivity and shows how corporate communications
can overinterpret simple associations.
```

Coverage analysis

The v0.2 expansion pack adds eight new cases that address gaps observed in the pilot set:

- **Domain diversification** – Several cases expand beyond health and supplements: education and AI (Case021 and Case025), workplace economics (Case022 and Case028), environment and public health (Case023 and Case026), and longevity gene therapy (Case024). Two synthetic cases bring in technology and productivity claims.
- **Overclaim strength** – Case025 offers a clear media overclaim about AI replacing teachers, while Case027 provides a synthetic but sharp example of unsupported technology hype. Case026 also shows how advocacy can overstate observational findings.
- **Correlation traps** – Case023 and Case028 reinforce correlation-vs-causation patterns in environmental and workplace contexts. Case026 blends advocacy and correlation, testing borderline recognition.
- **Mechanism versus evidence** – Case024 provides a robust mechanism description (Klotho gene therapy) with animal evidence but no human data, highlighting the mechanism_not_demonstrated failure mode. Case027 also includes pseudo-mechanistic claims.
- **Borderline ambiguity** – Cases 022 and 026 are explicitly labelled borderline to check whether Agent Arena can handle nuance when evidence is not strong but claims are somewhat restrained.

Remaining gaps: Even with v0.2, some areas remain under-represented. Future packs could include more cases from economics (e.g., minimum wage studies), public policy evaluations (e.g., universal basic income pilots), and technology ethics (e.g., algorithmic bias claims). Additional mechanism-heavy cases could come from biomedical research where mechanisms are plausible but evidence is

preliminary. It would also be valuable to include more strong negative examples (clearly no failure) from non-health domains.
